

TECHNICAL VALIDATION & HANDOFF

Heston and Double-Heston Models

Leakage, overfitting, validity, accuracy, parameter and continuation reference

Prepared for: Power-options project team

Version date: 06 August 2026

Scope: Single Heston and Double Heston only

Evidence status: VALIDATED_WITH_DISCLOSED_LIMITATIONS — 60/60 independent checks passed

Locked input: a8a56dd7b17074d8fa32f88936c8b404456f22f296d9dc4665faf3b13621f1d3

Decision Keep the current locked Single Heston as the historical winner. Do not promote the present Double Heston. This is a reproducibility-qualified conclusion, not a guarantee of future performance.

Critical honesty boundary The historical test period was previously viewed during earlier development, and next-session surface scores use realized target spot plus listed target strike/expiry coordinates. Future locked NSE dates are required for pristine forward confirmation.

1. Executive verdict

The audit found no critical inconsistency in data lineage, chronological splitting, candidate selection, state propagation, pricing, implied-volatility inversion, no-arbitrage bounds, prediction coverage, or saved metrics. All 60 independent checks passed, and all five existing automated test modules exited successfully.

Question	Evidence-backed answer
Was test data used to fit or select the saved candidates?	No row-level evidence of that pathway was found. Candidates are train-only; selection is validation-only.
Is zero overfitting proven?	No. A finite historical experiment cannot prove zero overfitting, especially after the test graph has been viewed.
Which model wins historically?	Single Heston. On 8,814 identical finite rows: RMSE 0.044200 vs 0.045510 for Double.
Should Double Heston be used now?	Not as the preferred model. It loses overall, wins only 2/11 stocks, and all 55 candidates are boundary-near.
Is this a pure unseen-day forecast?	No. It forecasts the conditional volatility surface using realized target spot and observed target contract coordinates.

Source: Independent recomputation in `outputs/heston\_handoff/independent\_heston\_audit\_checks.csv`.

2. Scope, exclusions and authenticity

This handoff covers only Single Heston and Double Heston. It does not begin neural-network training, PINN development, data augmentation, Kaggle replacement, PostgreSQL mutation, or a new NSE download.

Item	Verified value
Model-input file	outputs/019fc8a0/model_input_option_prices.csv
SHA-256	a8a56dd7b17074d8fa32f88936c8b404456f22f296d9dc4665faf3b13621f1d3
Rows in clean release file	572,512
Rows flagged model-ready	215,636
Stocks in model-ready subset	11
Original row key	symbol trade date exact expiry option type strike
Original source files	raw/nse_fo_bhavcopies/<year>/...csv.zip
Missing referenced raw files	0

Authenticity means the evaluated rows resolve to local raw NSE FO bhavcopy paths and the locked file hash. This audit does not independently certify NSE's exchange infrastructure or redownload every bhavcopy from the internet.

3. Chronology and information boundary

- Dates are split 70% train, 15% validation and 15% test independently within each stock.
- All Single- and Double-Heston candidate training dates precede the first validation date for the same stock.
- Candidate selection is recomputable using validation data only; validation and test row keys are disjoint.
- Every saved origin date strictly precedes its target date; the observed calendar gap is one to four days.
- Exact expiry_date and days-to-expiry values are preserved. Historical Tuesday, Wednesday and Thursday expiries remain distinct; no weekday rule rewrites them.
- Origin-state tables contain no target option price or target implied-volatility label.

What 'no leakage found' means The saved row-level artifacts reproduce a valid train → validation selection → test scoring path. It does not prove that a researcher never looked at aggregated test results. The project record explicitly says the test graph was viewed earlier.

4. Implemented model equations and parameter meaning

4.1 Single Heston

Under the risk-neutral measure, the implemented one-factor model uses stochastic variance $v(t)$:

Price process $dS(t)/S(t) = (r - q)dt + \sqrt{v(t)} dW_S(t)$

Variance process $dv(t) = \kappa[\theta - v(t)]dt + \sigma\sqrt{v(t)} dW_v(t)$, with $d\langle W_S, W_v \rangle = \rho dt$

Parameter	Meaning	Implementation role
κ (kappa)	Mean-reversion speed	Propagates origin variance toward θ and shapes the characteristic function.
θ (theta)	Long-run variance	Positive structural level; annualized volatility scale is $\sqrt{\theta}$.
σ (sigma)	Volatility of variance	Controls smile curvature and variance uncertainty.
ρ (rho)	Spot/variance correlation	Controls skew; constrained inside $(-0.98, 0.98)$.
v_0	Date-specific variance state	Fitted at origin, propagated to target. Training median is descriptive, not the live forecast state.

The Feller diagnostic is $2\kappa\theta - \sigma^2 > 0$. Pricing uses a Little-Heston-trap characteristic function with 64-point Gauss-Laguerre integration, put-call parity for puts, and Black-Scholes inversion for implied volatility.

4.2 Double Heston

The implemented Double Heston characteristic function is the product of two constrained Heston-factor characteristic functions, represented additively in log-characteristic exponents. The instantaneous variance is decomposed into slow and fast states.

Two variance factors $dv_i(t) = \kappa_i[\theta_i - v_i(t)]dt + \sigma_i\sqrt{v_i(t)}dW_{vi}(t)$, for $i \in \{\text{slow, fast}\}$; total state = $v_{\text{slow}} + v_{\text{fast}}$.

- Operational parameter count is ten: eight structural parameters plus v_{0_slow} and v_{0_fast} .
- Slow/fast ordering is enforced by $\kappa_{\text{slow}} < \kappa_{\text{fast}}$.
- Both factors must satisfy separate Feller conditions.
- The joint correlation disk is constrained by $\rho_{\text{slow}}^2 + \rho_{\text{fast}}^2 < 1$.
- Each factor state propagates exactly as $\theta_i + [v_i(\text{origin}) - \theta_i] \exp(-\kappa_i \Delta\text{calendar}/365)$.

5. Selected Single-Heston parameter set

Symbol	Candidate	κ	θ	σ	ρ	v0 median	Val RMSE
ADANIENSOL	half_a	9.996663	0.402712	2.822393	-0.173073	0.151467	0.049179
ADANIGREEN	half_a	3.981365	0.999666	2.806315	-0.074662	0.167640	0.077256
ADANIPOWER	main	9.996659	0.560397	3.039316	-0.055311	0.325296	0.023363
CESC	half_a	5.506302	0.879373	3.095346	0.134853	0.039588	0.073465
JSWENERGY	main	9.996663	0.426153	2.903374	-0.122050	0.098113	0.044995
NHPC	half_a	9.996663	0.258595	2.261676	0.012605	0.045768	0.042614
NTPC	half_a	9.996663	0.120830	1.545998	-0.094111	0.056204	0.048292
POWERGRID	main	9.996663	0.155460	1.753598	-0.114274	0.029378	0.046688
SJVN	main	9.996443	0.434051	2.930124	-0.001738	0.127590	0.044968
TATAPOWER	main	9.996663	0.175414	1.862741	0.030042	0.064200	0.052841
TORNTPOWER	half_b	9.996663	0.395235	2.796071	-0.113463	0.056940	0.050559

Source: Sanitized catalog `outputs/heston\_handoff/single\_heston\_selected\_parameter\_catalog.csv`.

Operational warning The v0 median above summarizes training states. For each forecast origin, use the origin-specific fitted v0 and propagate it; do not substitute the median as a universal live state.

6. Selected Double-Heston parameter set

6.1 Slow factor

Symbol	Candidate	κ slow	θ slow	σ slow	ρ slow	v_0 slow med
ADANIENSOL	training_half_a	4.755650	0.353849	1.821515	0.567525	0.045393
ADANIGREEN	training_half_a	3.813725	0.171570	1.137861	-0.949681	0.029927
ADANIPOWER	full_start_3	4.998340	0.845936	2.892500	-0.531453	0.078482
CESC	training_half_a	0.051704	0.150343	0.013593	-0.001108	0.005843
JSWENERGY	full_start_1	4.830192	0.336053	1.790856	0.650200	0.026803
NHPC	training_half_a	0.063574	0.135629	0.129923	-0.944830	0.004114
NTPC	training_half_b	4.997313	0.081178	0.895853	0.650635	0.022222
POWERGRID	full_start_3	4.995617	0.153755	1.232828	-0.735950	0.011819
SJVN	full_start_1	1.063320	0.150794	0.563266	-0.949680	0.016240
TATAPOWER	full_start_1	0.051665	0.030971	0.056253	-0.948730	0.006149
TORNTPOWER	full_start_3	4.850334	0.254662	1.561242	0.641609	0.033945

6.2 Fast factor

Symbol	Candidate	κ fast	θ fast	σ fast	ρ fast	v_0 fast med
ADANIENSOL	training_half_a	9.917287	0.206907	2.013533	-0.760214	0.093458
ADANIGREEN	training_half_a	3.927227	0.827656	2.536069	-0.000001	0.095305
ADANIPOWER	full_start_3	9.997614	0.153734	1.208556	0.787016	0.209173
CESC	training_half_a	5.762406	0.847978	3.109475	-0.000000	0.024120
JSWENERGY	full_start_1	9.869913	0.235272	2.142738	-0.691422	0.047385
NHPC	training_half_a	3.360026	0.660608	2.095670	-0.000007	0.030068
NTPC	training_half_b	9.997454	0.066465	1.146657	-0.691616	0.030299
POWERGRID	full_start_3	5.113148	0.109849	1.054234	0.600222	0.013167
SJVN	full_start_1	4.253820	0.842980	2.663736	-0.000002	0.053298
TATAPOWER	full_start_1	9.996651	0.174263	1.856621	-0.000001	0.053749
TORNTPOWER	full_start_3	9.781598	0.209969	2.015034	-0.698837	0.039619

Source: Sanitized catalog `outputs/heston\_handoff/double\_heston\_selected\_parameter\_catalog.csv`.

Do not interpret these as uniquely identified economics All 55 Double-Heston candidates are boundary-near, and training-half/start-value fits disagree materially. Similar prices can be produced by different parameter combinations.

7. Calibration and selection protocol

Stage	Single Heston	Double Heston
Structural candidates	3 per stock: full train + alternating train halves A/B	5 per stock: 3 full-train starts + train halves A/B
Training objective	Vega-scaled robust price residuals, equal date weighting	Same principle; two variance states per training surface
Selection	Minimum validation pooled IV RMSE; alphabetical tie break	Minimum validation mean stock-session IV RMSE; pooled RMSE then name tie breaks
Test use	Scoring only after selection	Scoring only after selection; identical common rows with Single
Failure handling	Finite selected predictions	4 invalid test IV rows retained separately; no imputation

Optimization convergence alone is not evidence of unique recovery. The controlled tests show that prices can be recovered accurately even when structural parameters are materially displaced by small noise, especially for Double Heston.

8. Accuracy on the locked historical evaluation

Experiment	Model	Rows	IV RMSE	IV MAE	Bias	R ²
Original next-session	Single Heston	8,818	0.044469	0.028048	-0.004604	0.814899
Original next-session	Prior-session median IV	8,818	0.068079	0.042060	-0.031587	0.566172
Identical common rows	Single Heston	8,814	0.044200	0.027957	-0.004502	0.816578
Identical common rows	Double Heston	8,814	0.045510	0.028264	-0.005601	0.805544

The stock-session bootstrap interval for Double minus Single RMSE is [0.000607, 0.001993], entirely positive. Lower is better; this favors Single Heston.

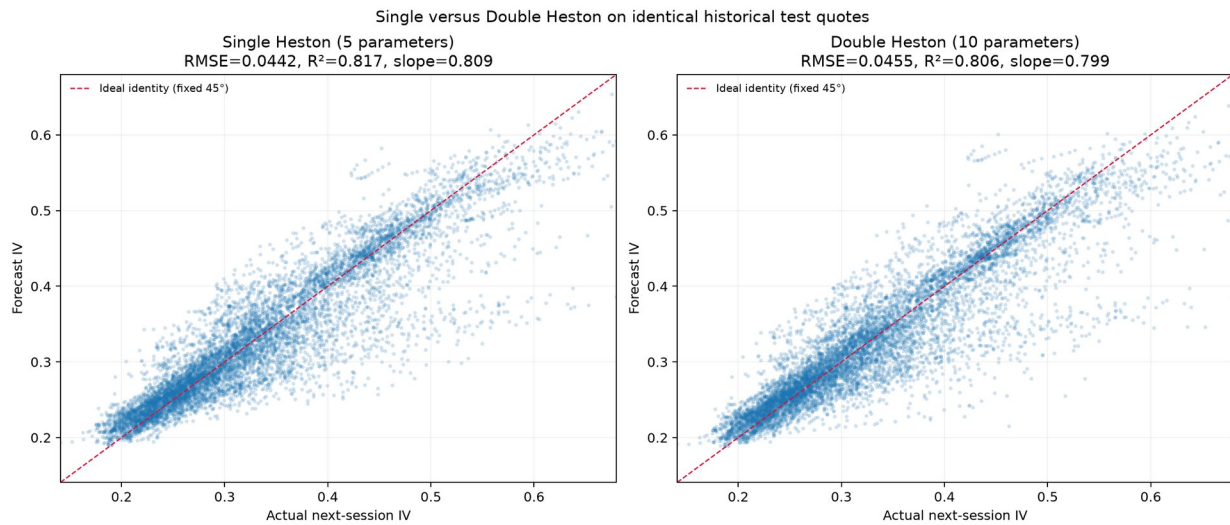


Figure 1. Identical-row actual IV versus Single- and Double-Heston forecasts.

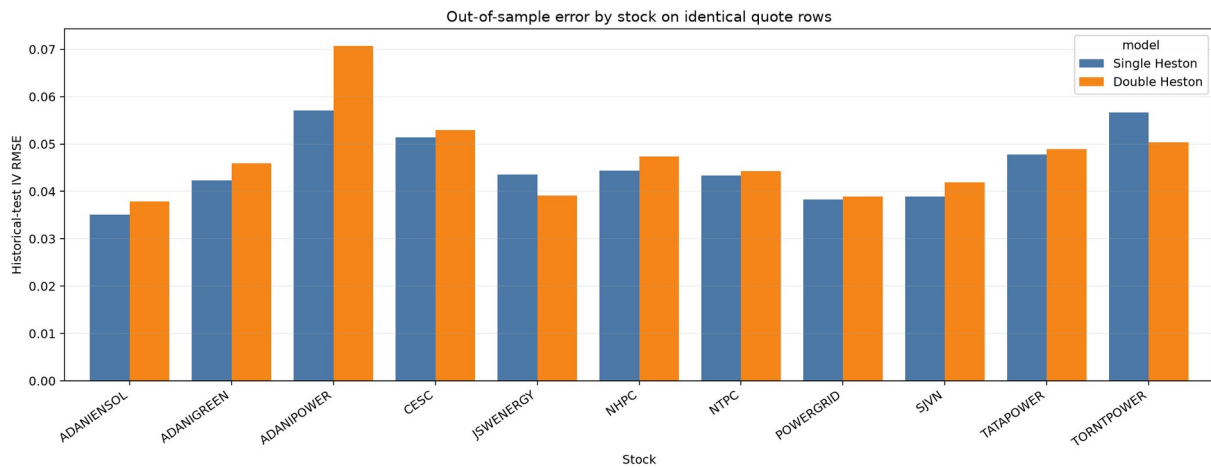


Figure 2. Per-stock test IV RMSE on the identical finite comparison universe.

8.1 Per-stock comparison

Symbol	Rows	Single RMSE	Double RMSE	Double-Single	Winner
ADANIENSOL	410	0.035058	0.037827	+0.002770	Single
ADANIGREEN	1336	0.042358	0.045958	+0.003599	Single
ADANIPOWER	183	0.057109	0.070736	+0.013627	Single
CESC	221	0.051420	0.052918	+0.001499	Single
JSWENERGY	596	0.043534	0.039108	-0.004426	Double
NHPC	605	0.044355	0.047366	+0.003012	Single
NTPC	1633	0.043370	0.044258	+0.000888	Single

Symbol	Rows	Single RMSE	Double RMSE	Double-Single	Winner
POWERGRID	1436	0.038244	0.038862	+0.000617	Single
SJVN	142	0.038933	0.041864	+0.002931	Single
TATAPOWER	1936	0.047752	0.048878	+0.001127	Single
TORNTPOWER	316	0.056687	0.050364	-0.006323	Double

Source: `outputs/single\_double\_heston\_comparison/single\_double\_test\_metrics.csv`.

9. Leakage and overfitting challenge tests

Challenge	Result	Interpretation
Three-fold strike crossfit	4,709 unique held-out quotes; Heston RMSE 0.030028	No same-row memorization in the sampled surfaces.
Within-surface centered R^2	0.617874	Less impressive than pooled R^2 0.903410; much pooled fit comes from same-day level calibration.
Flat same-day anchor	RMSE 0.040577	Heston beats a flat surface.
Quadratic same-day smile	RMSE 0.014712	Simple flexible smile beats Heston for same-day strike interpolation.
Prior available Heston state	RMSE 0.042245	Removing same-day state information reduces performance.
Shuffled state	RMSE 0.068991	Correct state timing matters.
Single half-sample fits	39 boundary flags across 22 fits	Structural parameters are not fully stable.
Double candidates	55/55 boundary-near	Severe identifiability/constraint-pressure warning.

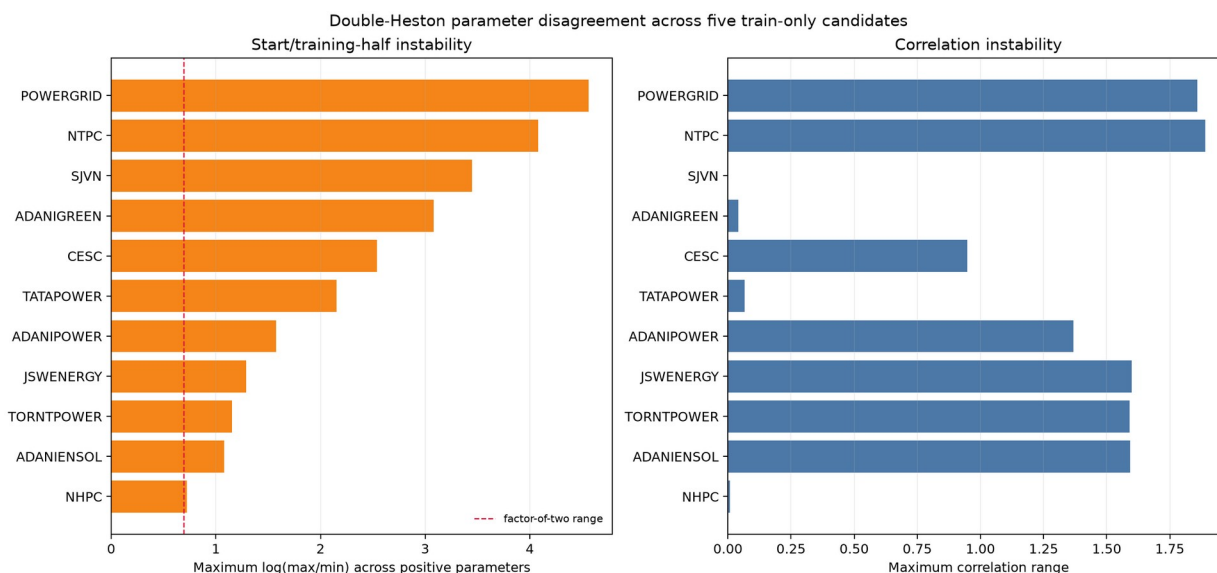


Figure 3. Double-Heston disagreement across five train-only candidates per stock.

The parameter-instability chart measures disagreement across optimizer starts and disjoint training halves. It is not a confidence interval and should not be interpreted as statistical uncertainty coverage.

10. Controlled synthetic recovery

Model / condition	Known-data recovery	Price error	Parameter sensitivity
Single, exact prices	Max relative error 3.222e-10	RMSE 2.817e-11	Three starts converge to effectively exact recovery
Single, 1% price noise	Recovered structural set remains close	RMSE 0.022329	Non-zero displacement; use price fit and stability together
Double, exact prices	Max structural error 1.189e-05	RMSE 3.679e-07	Implementation can recover a known clean solution
Double, 1% price noise	Max structural error 28.875%	RMSE 0.026107	Material non-identifiability despite small pricing loss

Meaning These tests validate pricing/calibration mechanics under controlled conditions. They do not validate that real NSE data are generated by Heston dynamics, nor do they make the historical parameter estimates unique.

11. Invalid rows, coverage and honesty controls

- The Single-Heston source test universe contains 8,818 unique rows.
- Double Heston returns finite IV for 8,814 rows (99.9546% coverage).

- Four invalid Double-Heston implied-volatility rows are retained in a separate failure file and are not imputed or silently replaced.
- Candidate selection uses only validation quote keys where all five Double candidates are finite; validation failures are also retained separately.
- All stored Single/Double prices reprice from the saved equations within 2×10^{-9} , and stored IV values invert those prices within 2×10^{-10} .

Symbol	Row key	Recorded result
NTPC	NTPC 2026-05-19 2026-05-26 CE 450.00	invalid IV
NTPC	NTPC 2026-06-19 2026-06-30 CE 460.00	invalid IV
NTPC	NTPC 2026-06-22 2026-06-30 CE 460.00	invalid IV
NTPC	NTPC 2026-07-21 2026-07-28 CE 460.00	invalid IV

Source: `outputs/single\_double\_heston\_comparison/double\_heston\_test\_prediction\_failures.csv`.

12. Retrospective generalization adjustment

A later validation-only guard improved the historical Single-Heston test RMSE from 0.044469 to 0.036590, with slope improving from 0.806598 to 0.878963. Corrections were accepted for NHPC, NTPC, POWERGRID, TATAPOWER.

Do not present this as pristine out-of-sample validation retrospective historical-test comparison; the original test graph was viewed before the improvement was designed, so future NSE sessions are required for pristine forward confirmation

For teammate continuation, the corrected model may be carried as a locked challenger, but its first credible confirmation must use dates that were not visible when the rule was designed.

13. Privacy and data-exposure review

Exposure class	Finding	Required action
Credentials/secrets	No API-key, password, AWS-key or private-key pattern found in reviewed model code/reports.	Continue secret scanning before any public release.
Personal/machine metadata	Two raw machine artifacts contain an absolute local path with the workstation username.	Do not share them verbatim; use sanitized catalogs and this scrubbed report.
Raw NSE rows	This report contains no raw option-price table and no embedded source dataset.	Share the data separately under the project's access policy.
Model parameters	All selected parameters are intentionally disclosed for	Treat them as project IP if required by team

Exposure class	Finding	Required action
	teammate continuation.	policy.

The final DOCX is metadata-scrubbed after generation. The context Markdown uses project-relative paths only.

14. Locked continuation protocol

1. Freeze `single\_heston.py`, `double\_heston.py`, `forecast\_single\_heston\_next\_day.py`, `compare\_single\_double\_heston.py`, selected parameter catalogs, and all checksum manifests.
2. Do not alter candidate starts, bounds, filters, correction thresholds, or evaluation rows using the previously viewed historical test.
3. Acquire authentic future NSE sessions strictly after 03 August 2026 and retain raw-source hashes and exact expiry dates.
4. Run the same model-ready filter and record every exclusion reason. Never fill or fabricate a failed model IV.
5. At each target, construct the origin state using origin-date information only; then propagate over the exact calendar gap.
6. Generate Single and Double predictions on identical row keys, conditional on the same target spot/contract coordinates if preserving the current evaluation convention.
7. Report pooled RMSE/MAE/bias/ R^2 , mean stock-session RMSE, forecast-on-actual slope, per-stock results, coverage and all failure rows.
8. Bootstrap by stock-session, not individual option row, because strikes within a session are dependent.
9. Promote Double only after a preregistered future window shows a clear improvement and parameter stability no longer depends strongly on start/half sample.

14.1 Recommended preregistration

Decision item	Lock before future scoring
Primary metric	Mean stock-session IV RMSE on identical finite rows
Uncertainty	95% stock-session cluster bootstrap of Double minus Single RMSE
Coverage floor	At least 99.9%; every invalid row documented
Promotion rule	Double interval entirely below zero and per-stock performance not concentrated in only one or two names
Stability rule	Substantially fewer boundary-near fits plus lower start/half-sample parameter ranges
No-retuning rule	One locked run on the preregistered future window

15. Reproducibility commands and file map

Run from the project root with the environment that provides NumPy, pandas, SciPy and Matplotlib:

```
python3 test_single_heston.py
python3 test_heston_overfit_audit.py
python3 test_heston_next_day_forecast.py
python3 test_heston_generalization.py
python3 test_double_heston.py
python3 audit_heston_handoff.py
```

Role	Project-relative path
Authentic model input	outputs/019fc8a0/model_input_option_prices.csv
Single core	single_heston.py
Single next-session	forecast_single_heston_next_day.py
Single overfit audit	audit_single_heston_overfitting.py
Double core	double_heston.py
Single/Double comparison	compare_single_double_heston.py
Independent handoff audit	audit_heston_handoff.py
Machine audit matrix	outputs/heston_handoff/independent_heston_audit_checks.csv
Sanitized parameters	outputs/heston_handoff/*_selected_parameter_catalog.csv
Context Markdown	outputs/heston_handoff/HESTON_DOUBLE_HESTON_TEAM_CONTEXT.md

16. Known limitations

- No finite historical test can prove zero overfitting or a global optimum.
- The historical test period was previously viewed during Single-Heston development; it is not pristine final holdout evidence.
- Next-session scoring is conditional on realized target spot and the target session's listed strike/expiry coordinates.
- Double Heston underperforms Single Heston on the identical historical rows and has severe parameter-boundary/identifiability warnings.
- The retrospective Single-Heston recalibration must be confirmed on future locked NSE dates before being called forward validated.

- The model universe is the 11 stocks with model-ready data, not every possible power-sector security.
- Calibration uses a filtered liquid/parity-consistent option subset; results do not automatically generalize to stale, illiquid or rejected quotes.
- Corporate-action adjustments and expiry changes are inherited from the locked clean-release pipeline; this audit validates the fields and lineage, not every corporate-action announcement independently.
- The baseline is useful but limited. Future work should retain it and may add stronger predeclared baselines without using the evaluation window to choose them.
- R^2 is descriptive and scale-sensitive; use RMSE, calibration slope, coverage, session clustering and stability together.

17. Audit-check appendix

Category	Independent check	Status	Observed
source	input sha256 matches locked NSE model input	PASS	a8a56dd7b17074d8fa32f88936c8b404456f22f296d9dc4665faf3b13621f1d3
source	model ready row count is locked	PASS	215636
source	model input row keys unique	PASS	0
source	model input contains 11 power stocks	PASS	11
source	model ready filter reproduces locked subset	PASS	all=572512, ready=215636
source	source paths are NSE bhavcopies	PASS	0
source	referenced raw NSE files exist locally	PASS	0
expiry	expiry distance matches saved days	PASS	0
expiry	maturity is exact days divided by 365	PASS	4.931506591976387e-11
expiry	multiple historical expiry weekdays preserved	PASS	Monday, Thursday, Tuesday
chronology	single candidates strictly precede validation	PASS	0
chronology	single origin strictly precedes target	PASS	0
chronology	single saved split matches independent 70 15 15	PASS	0
chronology	single validation and test keys disjoint	PASS	0
selection	single validation selection recomputed	PASS	11
selection	single selection declares validation only	PASS	0
selection	single has three train only candidates per stock	PASS	3
lineage	single test rows resolve to source	PASS	0

Category	Independent check	Status	Observed
lineage	single test values match source rows	PASS	0
equation	single variance propagation recomputed	PASS	1.1102230246251565e-16
equation	single saved prices reprice exactly	PASS	1.1368683772161603e-12
equation	single saved IV inverts saved prices	PASS	5.703770789011742e-13
validity	single prices within no arbitrage bounds	PASS	0
constraints	all single candidates Feller valid	PASS	0.02495313693270962
constraints	all single candidates within declared bounds	PASS	33 candidates
metrics	single selected heston metrics recomputed	PASS	9.367506770274758e-17
metrics	single prior session expiry median iv metrics recomputed	PASS	8.326672684688674e-17
chronology	double candidates strictly precede validation	PASS	0
selection	double has five train only candidates per stock	PASS	5
selection	each double validation key has all five candidates	PASS	5
selection	double validation selection recomputed	PASS	11
selection	double selection declares validation only	PASS	0
constraints	all double candidates satisfy both Feller conditions	PASS	3.28336959995474e-05
constraints	all double candidates have ordered factors	PASS	0
constraints	all double candidates satisfy joint correlation disk	PASS	0.9496814173760569
overfit	double boundary near candidates recorded	PASS	55
equation	double variance propagation recomputed	PASS	1.1102230246251565e-16
equation	double saved prices reprice exactly	PASS	6.821210263296962e-13
equation	double saved IV inverts saved prices	PASS	1.1843370728570335e-10
validity	double prices within no arbitrage bounds	PASS	0
coverage	single test universe has 8818 rows	PASS	8818
coverage	double common and failure rows partition single universe	PASS	common=8814, failures=4, universe=8818
coverage	double invalid rows are documented not imputed	PASS	4
lineage	single and double use identical common rows	PASS	8814
metrics	single heston comparison metrics recomputed	PASS	7.632783294297951e-17
metrics	double heston comparison metrics recomputed	PASS	8.326672684688674e-17
metrics	double minus single cluster bootstrap recomputed	PASS	0.0

Category	Independent check	Status	Observed
overfit	historical bootstrap favors single Heston	PASS	95% interval [0.00060695, 0.00199293]
overfit	single three fold strike crossfit metrics recomputed	PASS	3.469446951953614e-18
leakage	all strike crossfit leakage checks pass	PASS	11/11
overfit	quadratic same day smile beats single Heston	PASS	quadratic=0.014712; Heston=0.030028
overfit	single half sample boundary flags recorded	PASS	39
honesty	generalized single result labeled retrospective	PASS	retrospective historical-test comparison; the original test graph was viewed bef
honesty	conditional evaluation scope explicit	PASS	['CONDITIONAL_ON_REALIZED_TARGET_SPOT_AND_QUOTES']
controlled	single exact synthetic recovery	PASS	3.221947400236205e-10
controlled	single one percent noise sensitivity recorded	PASS	0.1241260944772804
controlled	double exact synthetic recovery	PASS	1.1886345312058081e-05
controlled	double one percent noise sensitivity recorded	PASS	0.28874634698904245
privacy	no credential patterns in reviewed code and reports	PASS	[]
privacy	raw machine artifacts with local paths identified	PASS	['outputs/ single_heston_next_day_forecast/ parameter_candidates_train_only.csv', ,

Source: Full details, severities and untruncated observations are in `independent\_heston\_audit\_checks.csv`.

18. Glossary

Term	Definition in this project
Origin	The last observed NSE session whose state is allowed to enter a target forecast.
Target	The next observed session being scored.
Conditional surface forecast	Forecast IV at target spot and target-listed strike/expiry coordinates, without using target option prices.
Structural parameters	κ , θ , σ and ρ for each variance factor.
State parameters	v_0 for Single; v_{0_slow} and v_{0_fast} for Double, fitted per origin session.
Feller gap	$2\kappa\theta - \sigma^2$; positive under the declared constraint.
Boundary-near	A diagnostic that a fitted candidate lies close to one or more imposed parameter limits.

Term	Definition in this project
Leakage	Information from validation/test labels influencing fitting, selection or state construction earlier than allowed.
Retrospective	Designed or interpreted after some test results were already viewed.

Handoff status The teammate can reproduce the saved experiments and continue with future locked validation. The report does not authorize calling the current results production-ready or leakage-proof in an absolute sense.